

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019-2020(Even Semester)
INNOVATIVE TEACHING PRACTICES

Degree, Semester& Branch: II Semester B.E. CSE

Course Code & Title: CS8251 Programming in C

Name of the Faculty member: Dr.K.Vijayalakshmi & Mrs.B.Vijayalakshmi

Innovative Practice's followed: Think-Write-Pair-Share

Type of Learning: Active Learning- Cooperative Learning

Date: 09.01.2020

Topic: Operators

Lecture No.: 5

Objectives of the activity:

- To enable the students to familiarize the concepts of various types of operators in C

Think-Write-Pair-Share

Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share) which was described in figure 1.

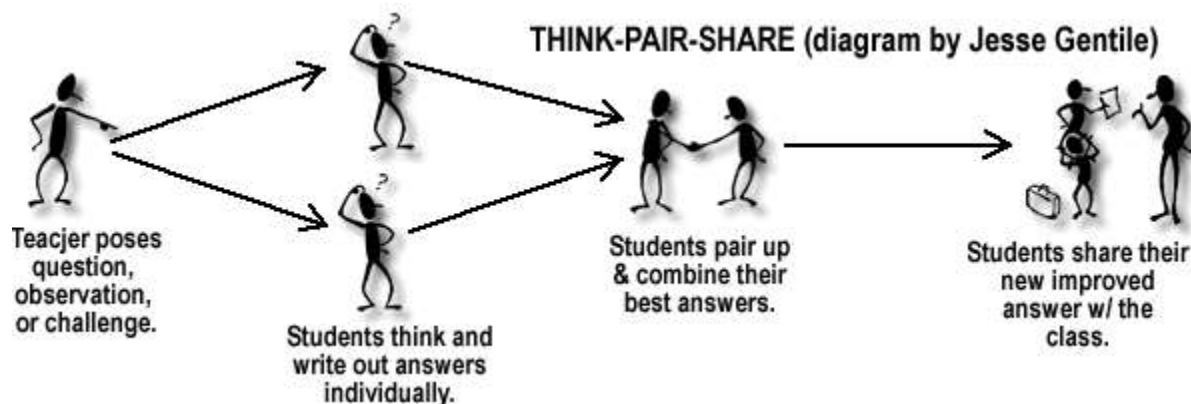


Figure 1: Think-Pair-Share Diagram

Use of think-pair-share

- It helps the students to think individually about a topic or answer to a question.
- It teaches the students to share ideas with classmates and builds oral communication skills.
- It helps focus attention and engage students in comprehending the reading material.

Justification for chosen the topic:

In C Programming, the concept of all operators plays a vital role. Hence the student must have a clear understanding of the various operators and its usage, precedence and associativity while executing an expression.

Implementation of TPS:

The activity was carried out in the classroom for the duration of 20 minutes.

- **Think - Write:**

Students were provided with a set of questions on operators for finding the output of the code, fill up the code with a proper operator to get the desired output, and find the errors as shown in the figure 2 & 3. The student has to think individually and write the answer in the given sheet. 10 minutes is allotted for this activity.

- **Pair:**

After finding the solutions to the given code, the students were allowed to discuss with neighbour and justify the resultant answer to each other as shown in figure 4.

- **Share:**

Once the conclusion is made between the pair, then it is to be shared among the whole class by individual student as shown in figure 5.

OUTCOME:

- The students were able to gain the knowledge and confidence in solving coding questions in C based on operators which will helpful in placement drives.
- The students were able to share their learning with others(peers).
- Apart from engaging students actively in learning, this activity improves retaining or remembering the operators for a long time

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Worksheet on Operators

1. What will be output of the following program?

```
#include <stdio.h>
void main() {
    int x;
    x=10,20,30;
    int y=(10,20,30);
    printf("%d",x);
    printf("%d",y);
    return 0;
}
```

Output: 10 30

2. If the output is $z=1$ then fill in the blanks with the suitable operator

```
#include <stdio.h>
int main()
{
    int x=12, y=7, z;
    z = x! = 4 | | y == 2;
    printf("z=%d\n", z);
    return 0;
}
```

3. What will be output of the following program?

```
int main()
{
    int i = -5;
    int k = i % 4;
    printf("%d", k);
    int x = 4.3 % 2;
    printf("Value of x is %d", x);
    return 0;
}
```

Output: -1 27001

4. Fill in the blanks to get the desired output:

```
#include <stdio.h>
void main()
{
    int a=202;
    int b=++a;
    int c=0++ ;
    printf("value of a = %d",a);
    printf("value of b = %d",b);
    printf("value of c = %d",c);
}
```

Figure 2: Questions on operators

Output:

Value of a = 204

Value of b=203

Value of c=203

5. Fill in the blanks to get the output of **b = 300 c = 200**

```
int main()
{
    int a = 500, b = 100, c;
    if(1a < 400)
        b = 300;
    c = 200;
    printf("b = %d c = %d\n", b, c);
    return 0;
}
```

6. Fill in the blanks to get the output as **x is small**

```
int main()
{
    int x=10, y=5;
    (x < y) ? printf("x is large"); printf("x is small");
    return 0;
}
```

7. What will be output of the following program?

```
#include <stdio.h>
int main()
{
    int i = 5, j = 10, k = 15;
    printf("%d", sizeof(k /= i + j));
    printf("%d", k);
    return 0;
}
```

Output: 2 15

8. Debug the code to get the desired output as **Value is 11**

```
#include <stdio.h>
void main()
{
    printf("Value is = %d", (10++));
}
```

Handwritten notes: ++10, int a = 10, a++

9. What will be output of the following program?

```
void main()
{
    int x = 4 * 5 / 2 + 9;
    printf("%d", x);
}
```

Handwritten note: output : 19

10. What will be the result of the following expressions if **i = 10** initially

- a. ++i <= 10
b. i++ <= 10

0
1

Figure 3: Questions on operators



Figure 4: Students discussion with neighbor.

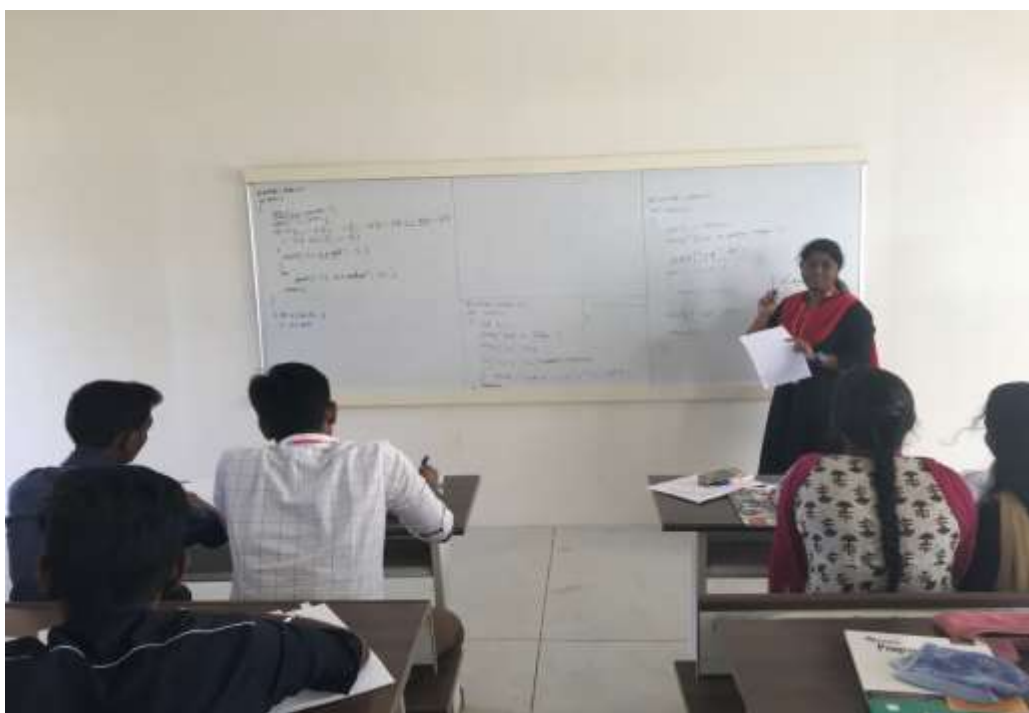


Figure 5: Sharing of knowledge

Relevance Mapping – This activity helps in attaining the following Objectives – PO mapping:

Objectives / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2	-	-	-	-	2	2	2	-	1

Reflective Report:**Pre-implementation:****Identified Challenges:**

- The activity needs more time to solve the problem individually and share their thoughts.
- Make all the students to involve actively in this activity.

Addressing the Challenges

- Discussed the importance of solving aptitude problems in C.
- Motivated them to involve actively in this activity which helps to improve their creative thinking.

Post-implementation:

- Coordination level among the students increased.
- Interest for solving coding problems gets improved
- All the students enjoyed this learning activity and builds confidence with students who are often hesitant to talk in front of other students.

References:

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6. <http://faculty.academyart.edu/faculty/teaching-topics/tips/think-write-pair-share.html>

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